

L2 - 9.5. Atomic Bird Goes Purple #1

Task 1 - Introduction

The module introduces the fundamentals of threat emulation, including key concepts, threat modeling for prioritizing risks, and practical use of tools like Atomic Red Team to simulate adversary techniques and evaluate defenses.

Invoke-AtomicTest is a PowerShell command from the **Atomic Red Team** project, used to execute predefined security tests that emulate real-world attacker behaviors mapped to the MITRE ATT&CK framework. It allows security teams to safely simulate adversary techniques in a controlled environment to:

- Validate security detections
- Test SIEM, EDR, and logging capabilities
- Assess incident response processes
- Verify security controls

Task 2 - Getting Started With Custom Exercises and Investigation Process

Threat Emulation and Custom Exercises: Threat Emulation involves simulating real-world attacker tactics, techniques, and procedures (TTPs) in a controlled environment. It helps organizations identify security gaps, improve detection capabilities, test security controls, and strengthen incident response readiness through realistic attack scenarios.

Task	Base Tactics	Reference Techniques	Implemented Actions
#4	• TA002: Execution • TA007: Discovery • TA009: Collection	• T1056.002 • T1059 • T1082	• System Enumeration • Input Capture • Command Execution
#5	• TA008: Lateral Movement	• T1091	• Data Manipulation on Shared Files
#6	• TA009: Collection	• T1115	• Data Dump • Clipboard and System File

Task	Base Tactics	Reference Techniques	Implemented Actions
			Modification

Investigation Process and Mindset

A well-configured endpoint will generate sufficient log files for threat emulation tests. Additional detection tools also increase visibility, and various options exist.

Note: Finding source code to analyse malicious files or attackers' tactics and techniques is not always possible. Experimental testing is one of the most common methods to overcome this challenge. Therefore, some atomics are not provided in cleartext, but task descriptions and event logs provide sufficient information to understand what to expect from each test.

Task 3 - Toolset and Hints

Toolset and Hints

- Windows Event Viewer
- Windows Registry Editor
- Custom Atomic Red Team Module
- THM-Utills Powershell module
- PowerShell

Atomic tests are based on PowerShell, so each time you execute a test or use a module, the system will generate a considerable amount of logs. You can clear the log files before each test so that the log files are not cluttered throughout the exercise.

THM-Utills usage

```

_____
|
| THM-Utills Commands |
| Result |
|_____|_____
|
| THM-LogClear-All -----> Clears all logs in
| the system. |

```

```

|   THM-LogStats-All -----> Application, Secur
ity, System, PowerShell,      |
|           |contd -----> PowerShell Operati
onal and Sysmon logs stats    |
|   THM-LogStats-Application -----> Summary of Applica
tion logs. (NO Aurora!)       |
|   THM-LogStats-Aurora -----> Summary of Aurora
agent logs.                   |
|   THM-LogStats-Flag -----> Gives the flag for
the question.                 |
|   THM-LogStats-PowerShell -----> Summary of PowerSh
ell logs.                     |
|   THM-LogStats-Powershell-Operational ---> Summary of PowerSh
ell Operational logs.         |
|   THM-LogStats-Security -----> Summary of Securit
y logs.                       |
|   THM-LogStats-Sysmon -----> Summary of Sysmon
logs.                         |
|   THM-LogStats-System -----> Summary of System
logs.                         |
|_____
|_____

```

Atomic Red Team usage

```

|_____
|_____
|   Atomic Red Team Command      |
Result                           |
|_____
|_____
|   help Invoke-AtomicTest -----> Shows th
e default help page.           |
|   Invoke-AtomicTest All -ShowDetailsBrief -----> Lists al
l tests.                       |
|   Invoke-AtomicTest T0000-1 -----> Executes
1st test case of the T0000-1 technique. |

```

```
| Invoke-AtomicTest T0000-1 -Cleanup -----> Removes  
the artefacts and restores the modified files (if any!). |  
|_____  
|_____
```

Note: The Powershell profile is customised to load required modules and provides atomic usage hints, as shown below. The THM-Utills module helps summarise important log files by grouping each event log based on four categories: Sum (Count), Event ID, Task Category and Event Provider.

```
Atomic Hint -> Show help: help Invoke-AtomicTest  
Atomic Hint -> List all tests: Invoke-AtomicTest All -ShowDetailsBrief  
Atomic Hint -> Execute test: Invoke-AtomicTest TXXX-1  
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup  
PS C:\Users\Administrator> THM-LogStats-All
```

Answer the questions below

Use the required PowerShell command to retrieve the flag. What is the flag?

THM{Emulation_is_fun_but_needs_focus_and_exploration} (Run the `thm-LogStats-Flag` executable in PowerShell.)

```
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup  
THM-Util Module Hint -> Cleanup Logs: THM-LogClear-All  
  
PS C:\Users\Administrator> THM-LogStats-Flag  
  
##|##|##| THM{Emulation_is_fun_but_needs_focus_and_exploration} |##|##|##|  
PS C:\Users\Administrator>
```

What is the required command to clear all generated artefacts and restore the affected files from test T0123-4?

Invoke-AtomicTest T0123-4 -Cleanup

Task 4 - Execute, Investigate, Detect

Reference Techniques	Given atomic tests are inspired by the following techniques. • T1082 System Information Discovery • T1056.002 Input Capture: GUI Input Capture
Storyline	Purple Team aims to simulate system discovery, GUI prompts and command execution. As a team member, your task is to discover the cleartext credentials, create accounts with given custom atomics and evaluate the generated artefacts.
Objective	Experiencing the impacts and artefacts of system discovery, user prompts and command execution actions.

The planned tests for this case are listed below.

```
T0004-1 TASK-4.1 Initial Enumeration Emulation
T0004-2 TASK-4.2 Credential Prompt Emulation
T0004-3 TASK-4.3 Failed command emulation
```

Note: You can revert the system modification and file change activities by using the cleanup command of the executed technique!

Answer the questions below

Execute test T0004-1 and open the document created on the Desktop. What is the OS Build info?

10.0.17763 N/A Build 17763 (After executing `Invoke-AtomicTest`, a TXT file containing OS information was downloaded.)

```
Administrator: Windows PowerShell

Atomic Hint -> Show help: help Invoke-AtomicTest
Atomic Hint -> List all tests: Invoke-AtomicTest All -ShowDetailsBrief
Atomic Hint -> Execute test: Invoke-AtomicTest TXXX-1
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup
THM-Util Module Hint -> Cleanup Logs: THM-LogClear-All

PS C:\Users\Administrator> THM-LogStats-Flag

|#|#|#|#| THM{Emulation_is_fun_but_needs_focus_and_exploration} |#|#|#|#|
PS C:\Users\Administrator> Invoke-AtomicTest T0004-1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

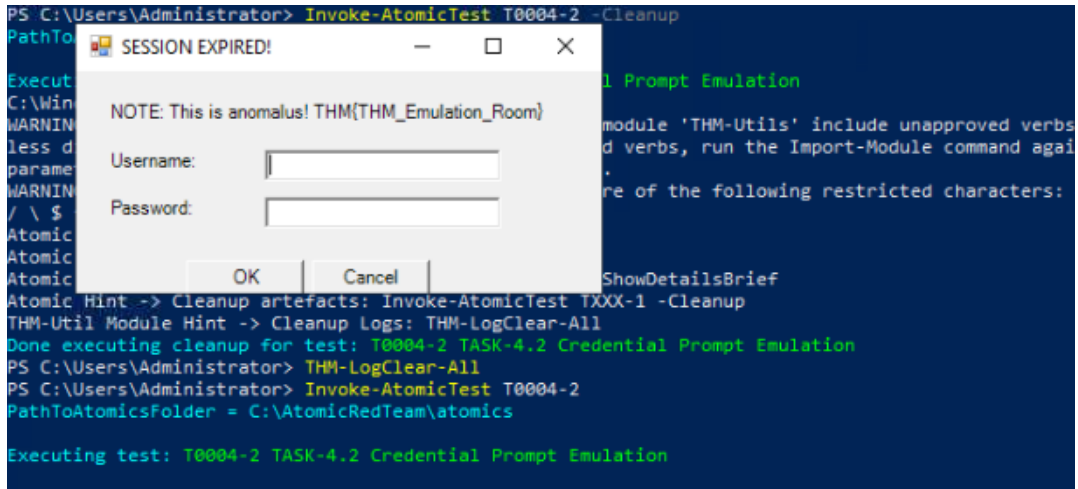
Executing test: T0004-1 TASK-4.1 Initial Enumeration Emulation
C:\Windows\System32\WindowsPowerShell\v1.0\profile.ps1
WARNING: The names of some imported commands from the module 'THM-Utils' include unapproved verbs discoverable. To find the commands with unapproved verbs, run the Import-Module command with the -UnapprovedVerbs parameter. For a list of approved verbs, type Get-Verb.
WARNING: Some imported command names contain one or more of the following restricted characters: / \ $ ^ ; : " ' < > | ? @ ` * % + = ~
Atomic Hint -> Show help: help Invoke-AtomicTest
Atomic Hint -> List all tests: Invoke-AtomicTest All -ShowDetailsBrief
Atomic Hint -> Execute test: Invoke-AtomicTest TXXX-1
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup
THM-Util Module Hint -> Cleanup Logs: THM-LogClear-All
Done executing test: T0004-1 TASK-4.1 Initial Enumeration Emulation
PS C:\Users\Administrator>
```

Task-Result.txt - Notepad

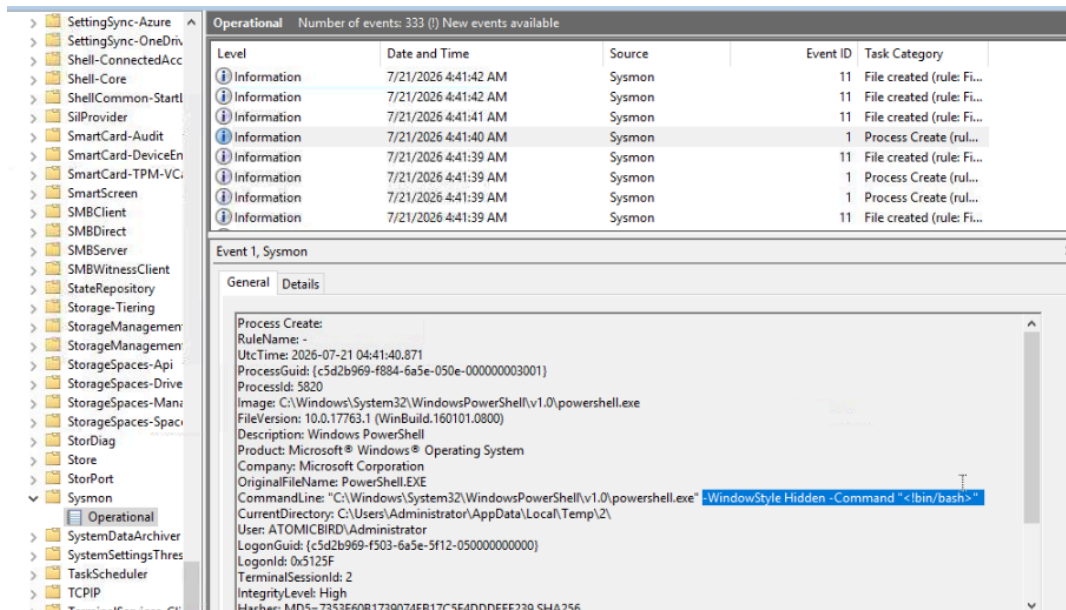
Host Name:	ATOMICBIRD
OS Name:	Microsoft Windows Server 2019 Datacenter
OS Version:	10.0.17763 N/A Build 17763
System Type:	x64-based PC
Hotfix(s):	27 Hotfix(s) Installed.

Execute test T0004-2. What is the flag?

THM{THM_Emulation_Room} (After executing `Invoke-AtomicTest`, a 'Session Expired' screen popped up)



Execute test T0004-3. Examine the logs; what is the failed command? <!bin/bash>



Task 5 - Universal Suspicious Share

Reference Techniques	Given atomic tests are inspired by the following techniques. T1091 Replication Through Removable Media
Storyline	Purple Team aims to simulate file manipulation actions on shared drives/files. As a team member, your task is to manipulate the shared files with given custom atomics and evaluate the generated artefacts.
Objective	Experiencing the potential impact of file manipulation, becoming comfortable with executing commands and analysing the

results

T0005-1 TASK-5 Universal Suspicious Share

Answer the questions below

Navigate the disk and drives, and open the shared folder. What is the SHA256 value of the ".txt" document?

3CA9FB42ACF0A347BDFDC78E0435331BC458194E4BC7FBFFB255BC4CF02CDC1A

```
PS C:\Users\Administrator> Get-FileHash 'S:\Donation_call.txt'
>>

Algorithm      Hash
-----
SHA256         3CA9FB42ACF0A347BDFDC78E0435331BC458194E4BC7FBFFB255BC4CF02CDC1A
Path
-----
S:\Donation_call.txt
```

Execute the test T0005-1. Re-calculate the SHA256 value of the document. What is the hash value?

626DBB861DCFF600DABEFCE7BF93F2C72C0F6462CC5729B963FC8242D7D43990

```
PS C:\Users\Administrator> Invoke-AtomicTest T0005-1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T0005-1 TASK-5 Universal Suspicious Share
C:\Windows\System32\WindowsPowerShell\v1.0\profile.ps1
WARNING: The names of some imported commands from the module 'THM-Utils' include unapproved verbs that might make them
less discoverable. To find the commands with unapproved verbs, run the Import-Module command again with the Verbose
parameter. For a list of approved verbs, type Get-Verb.
WARNING: Some imported command names contain one or more of the following restricted characters: # , ( ) { } [ ] & -
/ \ $ ^ ; : " ' < > | ? @ ` * % + = ~
Atomic Hint -> Show help: help Invoke-AtomicTest
Atomic Hint -> List all tests: Invoke-AtomicTest All -ShowDetailsBrief
Atomic Hint -> Execute test: Invoke-AtomicTest TXXX-1
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup
THM-Util Module Hint -> Cleanup Logs: THM-LogClear-All
Done executing test: T0005-1 TASK-5 Universal Suspicious Share
PS C:\Users\Administrator> Get-FileHash 'S:\Donation_call.txt'

Algorithm      Hash
-----
SHA256         626DBB861DCFF600DABEFCE7BF93F2C72C0F6462CC5729B963FC8242D7D43990
Path
-----
S:\Donation_call.txt

PS C:\Users\Administrator>
```

Task 6 - Dump and Go

Reference Techniques	Given atomic tests are inspired by the following techniques. • T1115 Clipboard Data
Storyline	Purple Team aims to simulate storing command-line history and hijacking system files for multiple aims, including Man-in-the-Middle (MITM), exfiltration, and trick security product workflow. As a team

	member, your task is to dump command-line history, hijack system files with given custom atomics and evaluate the generated artefacts.
Objective	Experiencing the potential impacts of command-line history dumps and system file hijacks.

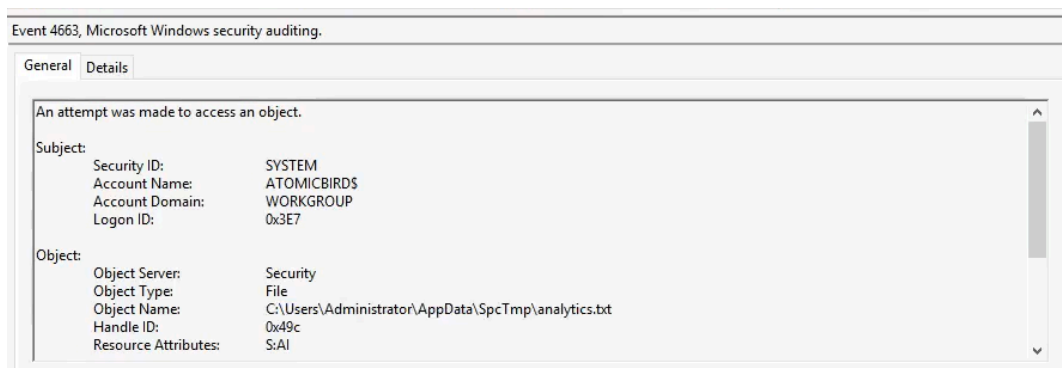
T0006-1 TASK-6.1 History dump

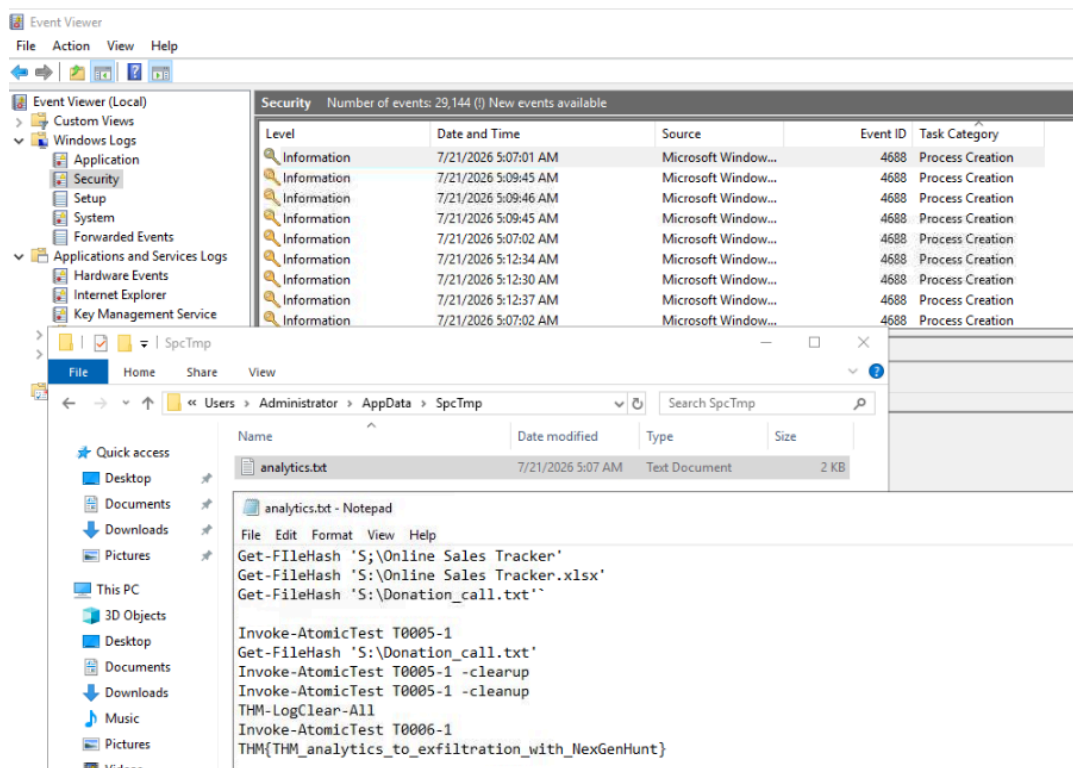
T0006-2 TASK-6.2 SystemFile modification for exfiltration

Answer the questions below

Execute test T0006-1. Find the malicious history dump file. What is the flag?

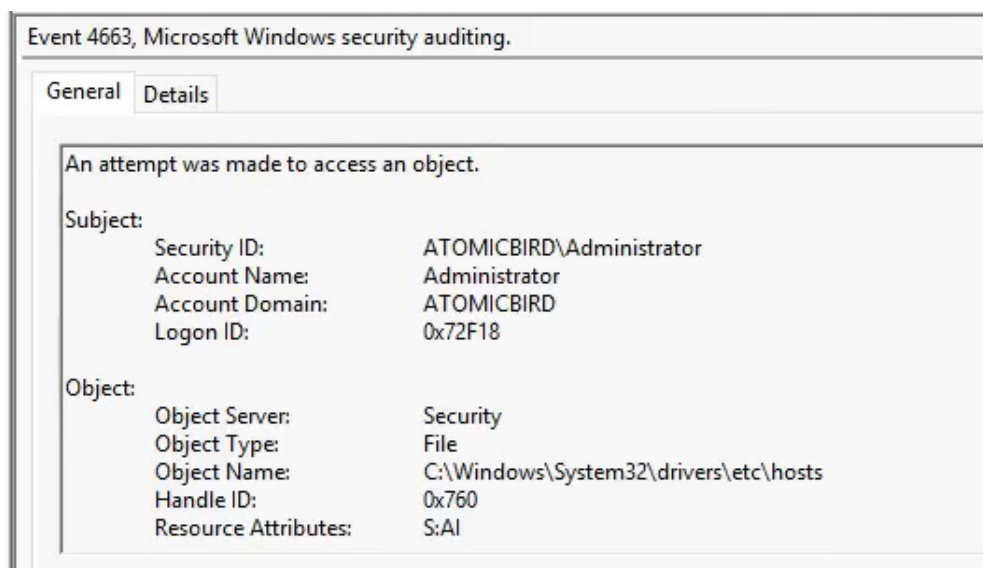
THM{THM_analytics_to_exfiltration_with_NexGenHunt} (Investigate why the attacker wants to access this file, then review the details of the next file that was created.)





Execute test T0006-2. Find the malicious system file modification activity. What is the flag?

THM{NextGenHunt.thm.jhn}



Level	Date and Time	Source	Event ID	Task Category
Information	7/21/2026 5:24:21 AM	Microsoft Window...	4658	File System
Information	7/21/2026 5:24:21 AM	Microsoft Window...	4658	File System
Information	7/21/2026 5:24:21 AM	Microsoft Window...	4658	File System
Information	7/21/2026 5:24:21 AM	Microsoft Window...	4656	File System
Information	7/21/2026 5:24:21 AM	Microsoft Window...	4658	File System
Information	7/21/2026 5:24:21 AM	Microsoft Window...	4656	File System

Local Disk (C:) > Windows > Sys

Name

- hosts
- lmhosts.sam
- networks
- protocol
- services

hosts - Notepad

```

File Edit Format View Help
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#      102.54.94.97      rhino.acme.com      # source server
#      38.25.63.10      x.acme.com         # x client host
#
# localhost name resolution is handled within DNS itself.
#
#      127.0.0.1        localhost
#      ::1              localhost
THM.10.10.JHN      THM{NextGenHunt.thm.jhn}      # NextGenHunt analytics

```

(Event ID 4663 shows that the Administrator account accessed the Windows hosts file (`C:\Windows\System32\drivers\etc\hosts`). The hosts file is a sensitive system file that can be abused for DNS redirection or blocking security services. Further investigation should focus on the process responsible for the access and whether the file was modified.)

Task 7 - Conclusion

In this room, we covered the implementation of custom atomic tests and detection for Purple Team, Threat Emulation and Detection Engineering exercises.